

Pavan Kumar Gannaju

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Professional Summary

Software Engineer with experience building scalable software systems, execution pipelines, and API-driven platforms using Python, FastAPI, and WebSockets. Skilled in backend engineering, system integration, monitoring, and real-time communication architectures, with strong interest in infrastructure engineering, distributed systems, Linux environments, containerization, and reliable software platforms.

Experience

Software Engineering Intern – Digital Nexus AI, Bangalore

May 2025 – Sep 2025 | Full-Time

- Developed scalable backend services and REST APIs using Python to support reliable inter-service communication across enterprise applications.
- Debugged and resolved backend **defects** through **server-log analysis** and **API** request tracing, improving application **stability** and reducing **recurring issues**.
- Performed **unit** and **functional** testing to validate backend **reliability**, **data integrity**, and deployment readiness.
- Collaborated within **Agile development teams** following **SDLC** practices and **Git-based** version control workflows.

Software Engineering Intern – Prodigal AI Technologies Pvt Ltd, Delhi

Mar 2025 – Nov 2025 | Remote Part-Time

- Coordinated development tasks across a **small engineering team** to support modular **backend** workflows and **API** integration tasks.
- Built and optimized modular data-processing pipelines in Python, improving workflow scalability and reducing response latency.
- Analyzed execution **logs** and **edge-case failures** to support **debugging**, **defect** resolution, and **reliability** improvements.
- Assisted in **backend testing**, **feature integration**, and **iterative software** delivery within Agile development environments.

Projects

Access Control & Authentication Gateway | Python, FastAPI, MySQL, REST APIs, JWT Authentication, Docker

- Engineered a secure access-control framework enforcing isolated authorization policies across **4 user roles**, improving permission management and access reliability.
- Integrated JWT authentication, audit logging, monitoring workflows, and API security mechanisms to improve request traceability and system reliability.
- Developed functional and security validation suites covering **50+** authorization scenarios, reducing unauthorized-access vulnerabilities and improving testing coverage.
- Optimized authentication workflows and SQL query execution, reducing authorization latency by **40%** under concurrent request conditions.

API Traffic Management & Rate-Limiting Engine | Python, FastAPI, PostgreSQL, REST APIs, Docker, Rate Limiting

- Built containerized middleware services for API traffic regulation and rate-limiting, improving scalability and request-handling stability under simulated high-load conditions.
- Implemented request **logging**, **monitoring**, and **validation** pipelines to analyze traffic behavior and identify backend performance bottlenecks.
- Designed request-filtering and validation mechanisms that reduced malformed-request failures by **90%** and improved API reliability.
- Performed load and stress testing under **3×** expected traffic conditions in **containerized** Linux-based environments to validate backend stability and service performance.

Skills

- Programming Languages:** Python, Java, C++, SQL
- Backend & Infrastructure:** FastAPI, REST APIs, WebSockets, Linux, Docker, Containerization, SQLite
- Databases & Tools:** MySQL, PostgreSQL, MongoDB, Git, GitHub Actions, Postman
- Software Engineering:** SDLC, Agile, Debugging, Root Cause Analysis, Unit Testing, API Testing, Monitoring
- Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Distributed Systems

Education

- Bachelor of Technology in Computer Science & Engineering (AI/ML)**
Visvesvaraya College of Engineering and Technology | Sep 2023 – May 2026 | Lateral Entry | CGPA: **8.01/10.0**
- Indian Institute of Technology, Ropar** – Minor in Artificial Intelligence | Sep 2024 – Sep 2025

Leadership & Responsibilities

- Organizer & Mentor — Code for Fun & Neuro Quest Hackathons:** Organized and managed two university-level hackathons involving **50+** participants by coordinating technical activities, debugging sessions, and algorithmic problem-solving mentorship.
- Team Lead — Academic Project Team:** Led a **4-member** development team by coordinating implementation tasks, conducting code reviews, and maintaining coding standards across project modules.